IB Mathematics Application & Interpretation
Higher Level–Prediction Exams

November 2026 - Paper 3

Duration: 1:15 hrs **(Calculator is Allowed)** **Max. Marks: 55**

Question 1

[Maximum mark: 27]

This question asks you to investigate the road network connecting five towns and the delivery times along a chosen route.

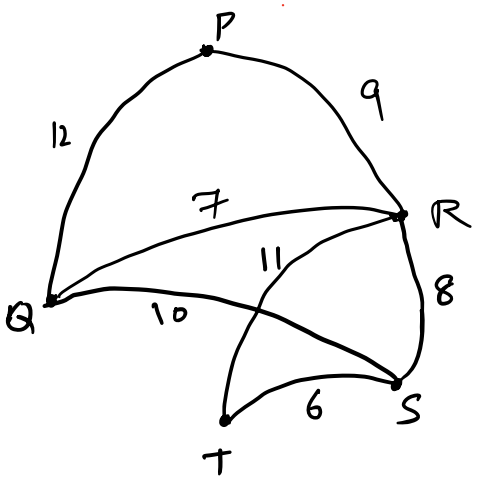
A delivery company operates in a region containing five towns P, Q, R, S and T. The direct roads between the towns, with their lengths in kilometres, are shown in the table below. A dash $(-)$ indicates that there is no direct road between the two towns.

	P	Q	R	S	T
P	–	12	9	–	–
Q	12	–	7	10	–
R	9	7	–	8	11
S	–	10	8	–	6
T	–	–	11	6	–

(a) Sketch a weighted graph representing the delivery region. [1]

We have $PQ=12, PR=9, QR=7, QS=10, RS=8, RT=11, ST=6$

Following is possible solution.



A delivery vehicle is to travel along every road in the network exactly once.

(b) Determine whether this is possible. Justify your answer. [2]

Answer:

To determine whether a driver can travel every road exactly once, we need to find degree of each vertex:

$$\deg(P) = 2$$
$$\deg(Q) = 3$$
$$\deg(R) = 4$$
$$\deg(S) = 3$$
$$\deg(T) = 2$$

Since exactly two vertices Q & S have odd degree, an Eulerian trail exists. So it is possible to travel along every road in network once, starting & finishing at vertices Q & S.

Yes, since exactly two vertices have odd degree.

(c) Use Kruskal's algorithm to find a minimum spanning tree for the network, and state its total length. [3]

Answer:

According to Kruskal's algorithm, we:

- Find the edge of least weight anywhere in the graph as our first edge,
- Continue to add the next edge of least weight from anywhere in the graph that does not form a cycle;
- Repeat until all vertices are connected.

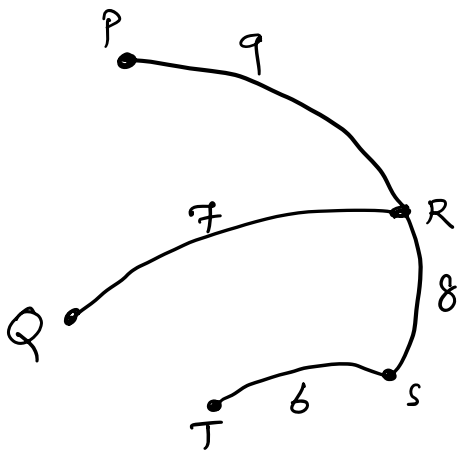
The edges are:

$ST=6, QR=7, RS=8, PR=9, QS=10, RT=11, PQ=12$

Adding edges in this order,

Hence total length of minimum spanning tree is:

$$6+7+8+9 = 30km$$



(d) An inspector must travel along every road at least once, starting and finishing at town P. Find the length of the shortest possible route. [4]

Answer:

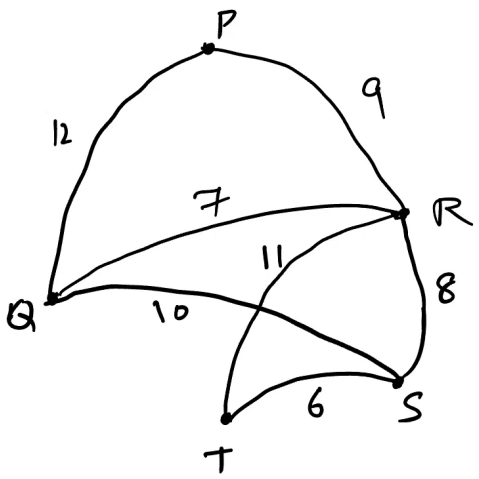
$$12 + 9 + 7 + 10 + 8 + 11 + 6 = 63$$

From part(b), the vertices Q & S have odd degree, so the shortest path between Q & S must be repeated to allow a route that starts & finishes at the same town.
Comparing possible routes between Q & S, we have

$$QS = 10, QRS = 7 + 8 = 15$$

Hence length of shortest possible route is,

$$63 + 10 = 73km$$



$12+7+8+10+10+6+11+9=73$

The company now wishes to find a route that starts at P, visits every town exactly once and returns to P. The table of least distances between each pair of towns is shown below.

	P	Q	R	S	T
P	–	12	9	17	20
Q	12	–	7	10	16
R	9	7	–	8	11
S	17	10	8	–	6
T	20	16	11	6	–

(e) Use the nearest-neighbour algorithm, starting at town P, to find an upper bound for the length of such a route. [3]

Answer:

Applying the nearest neighbourhood algorithm from P, we have

$$P \to R\,(9) \qquad R \to Q\,(7) \qquad Q \to S\,(10) \qquad S \to T\,(6) \qquad T \to P\,(20)$$

So the route obtained is,

$$P \to R \to Q \to S \to T \to P$$

Hence upper bound for the length of the route is

$$9 + 7 + 10 + 6 + 20 = 52km$$

	P	Q	R	S	T
P	–	12	9	17	20
Q	12	–	7	10	16
R	9	7	–	8	11
S	17	10	8	–	6
T	20	16	11	6	–

(f) By deleting town P, find a lower bound for the length of such a route. [3]

	P	Q	R	S	T
P	–	12	9	17	20
Q	12	–	7	10	16
R	9	7	–	8	11
S	17	10	8	–	6
T	20	16	11	6	–

Deleting row & column of P and applying Kruskal's algorithm to the remaining vertices Q, R, S, T, the minimum spanning tree consists of edges:

ST = 6, QR = 7, RS = 8,

Total = 6 + 7 + 8 = 21

The two edges of least length incident to P are:

PR = 9, PQ = 12

∴ Lower bound for length of route is

$$21 + 9 + 12 = 42km$$

A delivery driver, John, follows the upper bound found in part (e), and stops for 5 minutes in every town to make a delivery. He completes this route 5 days a week, 48 weeks per year. It can be assumed that John's total time to complete the route each day is independent.

The speed at which John travels, X km h^{-1}, is normally distributed with a mean of 60 km h^{-1} and a standard deviation of 4 km h^{-1}.

His expected driving time in **hours**, while in motion, is given by

$$\mathrm{E(time)} = \frac{\text{distance travelled}}{\mathrm{E(speed)}}$$

And the variance in **hours^2** is

$$\mathrm{Var(time)} = \left(\frac{\text{distance travelled}}{(\mathrm{E(speed)})^2}\right)^2 \times \mathrm{Var(speed)}$$

It is given that the time taken for John to complete his route, Y **minutes**, is normally distributed.

(g) Find the distribution of Y, stating its parameters. [5]

(h) Find $\mathrm{P}(Y < 80)$. [2]

(g) $\mathrm{E(time)} = \frac{52}{60}$ hours

$= \frac{52}{60} \times 60$

$\therefore \mathrm{E(time)} = 52$ minutes

$\mathrm{Var(time)} = \left(\frac{52}{(60)^2}\right)^2 \times 4^2$ hours^2

$= \left(\frac{52}{(60)^2}\right)^2 \times 4^2 \times 60^2$ minutes^2

$\mathrm{Var(time)} = 12.01$ minutes^2

Using linear transformation of a random variable (4.14) , since John stops for 5×5 =25 minutes, we have

$\mathrm{E}(Y) = 52 + 25 = 77$

$\mathrm{Var}(Y) = \mathrm{Var(time)} = 12.01$ $\left\{\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)\right.$

Hence $Y \sim N(77, 12.0)$

(h) Using GDC, with distribution found in part(g), we have

$P(Y<80) = 0.807$ (s.d. $= \sqrt{12.01\ldots}$)

Over the course of a year, a delivery driver earns a bonus if they complete at least 80% of their delivery routes in under 80 minutes.

(i) Find the probability that John earns the bonus in the forthcoming year. [4]

The number of routes John completes in a year is $5 \times 48 = 240$

John earns bonous if he completes at least (80%)

$240 \times 0.80 = 192$

routes in under 80 minutes

Let N be the number of routes out of 240 the John completes in under so minutes.
This fits in Binomial distribution

- We have fixed number of trials (240 delivery days)
- Each trial is independent (given)
- Each trial has two outcomes (route is either finished in under 80 minutes, or it is not)
- There is constant probability of success(from part(h) , P(Y<80=0.8065...)

$$\therefore N \sim B(240, 0.8065\ldots)$$

Using GDC, we have

$P(N \geq 192) \approx 0.639$

Question 2

[Maximum mark: 28]

This question asks you to investigate a population model for the population of a fish species, and the impact when a new species is introduced.

A population of walleye fish in a lake is modelled by the differential equation

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{w(20-w)}{4(10-w)}, \quad 0 < w < 10$$

where w is the number of walleye, in thousands, and t is time in decades. Initially there are 2 000 walleye.

(a) Determine the initial growth rate of the walleye population, giving your answer as a whole number of fish per year. [3]

$$\frac{dw}{dt}|_{t=0} = \frac{2\,(20-2)}{4\,(10-2)} = 1.125$$

Converting it to rate per year,

$$1.125 \times \frac{1000}{10} = 112.5 \approx 113 \quad \text{(to the nearest integer)}$$

Initial growth rate of walleye population is 113 fish per year.

(b)

$$\frac{dw}{dt} = \frac{w\,(20-w)}{4\,(10-w)}$$

$$\int \frac{4\,(10-W)}{W\,(20-W)}dW = \int dt$$

Let $$\frac{4\,(10-w)}{w\,(20-w)} = \frac{A}{w} + \frac{B}{20-w}$$

$$\therefore 4\,(10-w) = A\,(20-w) + Bw$$

For $w = 0 \rightarrow 40 = 20A \rightarrow A = 2$

& For $w = 20 \rightarrow -40 = 20B \rightarrow B = -2$

$$\therefore \int \left(\frac{2}{w} - \frac{2}{20-w}\right) dw = \int 1 dt$$

$$2\ln w + 2\ln(20-w) = t + c$$

$$\ln W + \ln(20-W) = \frac{t}{2} + \frac{c}{2}$$

$$\ln|w(20-w)| = \frac{t}{2} + \frac{c}{2}$$

$$\therefore w(20-w) = e^{\frac{t}{2}+\frac{c}{2}}$$

$$= e^{\frac{t}{2}} \cdot e^{\frac{c}{2}}$$

$$w(20-w) = K \cdot e^{\frac{t}{2}}$$

where $K = e^{\frac{c}{2}}$ and $w(20-w) > 0$ since $0 < w < 10$

So absolute value is not required.

Initially, $w = 2$ when $t = 0$

$$\therefore 2(20-2) = ke^0$$

$$k = 36$$

$$w(20-w) = 36e^{\frac{t}{2}}$$

$$\therefore \quad 20w - w^2 = 36e^{\frac{t}{2}}$$

$$w^2 - 20w + 100 = 100 - 36e^{\frac{t}{2}}$$

$$(w-10)^2 = 100 - 36e^{\frac{t}{2}}$$

Now $0 < w < 10$ so $w - 10 < 0$

Taking negative square root, we have

$$w = 10 - \sqrt{100 - 36e^{\frac{t}{2}}}$$

(c) Explain why the model is only valid for 20 whole years. [2]

For w to be real, we require

$$100 - 36e^{\frac{t}{2}} \geqslant 0$$

$$100 \geqslant 36e^{\frac{t}{2}}$$

$$e^{\frac{t}{2}} \leqslant \frac{100}{36}$$

Solving for t, we have

$$t \leqslant 2.043\ldots \text{ decades}$$

or $t \leqslant 20.43\ldots$ years

As this lies between 20 and 21 years, the model remains valid through first 20 whole years.

(d) Hence find the number of walleye after 20 years. [2]

Since t is measured in decades, 20 years corresponds to $t = 2$.

From part (b)

$$W(2) = 10 - \sqrt{100 - 36e^1}$$

$$\therefore w(2) = 8.536\ldots$$

Since w is measured in thousands, the number of walleye after 20 years is

$$8.536\ldots \times 1000 = 8540 \text{ walleyes}$$

After 20 years, a population of 2 750 yellow perch are introduced to the lake, which affects the equilibrium of the walleye population. The population of both walleye and yellow perch are now modelled by the coupled system

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{w(20-w)}{40} - 10y, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{y^2}{10} - w,$$

where w is the number of walleye and y is the number of yellow perch, both in thousands, and t is time in decades.

(e) Using Euler's method for coupled systems with a step size of 0.1, find the population of both walleye and perch after 5 years. [5]

Using Euler's method for coupled system (5.16), we have

$$x_{n+1} = x_n + h \cdot f_1(x_n, y_n, t_n)$$
$$y_{n+1} = y_n + h \cdot f_2(x_n, y_n, t_n)$$
$$t_{n+1} = t_n + h$$

Substituting w and y in place of x and y, h = 01, and $f_1(x_n, y_n, t_n) = \frac{dw}{dt}|_n$ and $f_2(x_n, y_n, t_n) = \frac{dy}{dt}|_n$, we have

$$w_{n+1} = w_n + 0.1\left(\frac{W_n(20-w_n)}{40} - 10y_n\right)$$
$$y_{n+1} = y_n + 0.1\left(\frac{y_n^2}{10} - w_n\right)$$
$$t_{n+1} = t_n + 0.1$$

Using unrounded value of w(2) found in part(d) as $w_0 = 8.536492509$, $y_0 = 2.75, t_0 = 0$

So that 5 steps are needed to reach t = 0.5 decades (5 years)

n	t_n	w_n	y_n
0	0	8.53649...	2.75
1	0.1	6.03114	1.97198
2	0.2	4.26978	1.40775
3	0.3	3.02995	1.00059
4	0.4	2.1579	0.707605
5	0.5	1.54655	0.496822

y_n column C(w)
$= C_1 + 0.1 \times \left(C_1 \times \frac{(20-C_1)}{40} - 10 \times d_1\right)$
y_n column D(Y)
$= d_1 + 0.1 \times \left(\frac{d_1^2}{10} - C_1\right)$

So, after 5 years, the populations of walleye and yellow perch are

$$w = 1.54655 \times 1000 \simeq 1550$$

$$y = 0.496822 \times 1000 = 497$$

(f) Explain what happens to both species of fish over a long period of time. [2]

Continuing the iteration from part (e) , the yellow perch population keeps declining and reaches zero (becomes extinct) between t = 1 (10 years) and t = 1.1 (11 years). By this point walleye population has fallen to approximately 440.

Since the yellow perch population is extinct, we have over a long period of time, $y \longrightarrow 0$

Once the yellow perch population reaches zero , the walleye population evolves according to:

$$\frac{dw}{dt} = \frac{w\,(20 - w)}{40}$$ Which has stable equilibrium at w = 20, despite falling substantially

In the short term, in the long term, $w \longrightarrow 20$

Hence, in the long term, the yellow perch dies out completely, while the walleye population recovers and stabilizes at 20,000.

(g) Describe the long term effects on both populations if 2 800 yellow perch were introduced after 20 years instead of 2 750. [2]

Replacing $y_0 = 2.8$ instead of $y_0 = 2.75$

n	t_n	w_n	y_n
0	0	8.536...	2.8
1	0.1	5.981...	2.025...
2	0.2	4.166...	1.468...
3	0.3	2.863...	1.073...
4	0.4	1.913...	0.7977...
5	0.5	1.202...	0.6128...
6	0.6	0.6459...	0.4963...
7	0.7	0.1808...	0.4342...
8	0.8	−0.244...	0.4180...

Both populations again decline first, but this time it is walleye population that reaches zero first, becoming extinct between t = 0.7 (7 years) and t = 0.8 (8 years).
Once walleye population reaches zero, the yellow perch population evolves according to:

$$\frac{dy}{dt} = \frac{y^2}{10}$$ Which is positive for y > 0 and has no finite equilibrium.

The perch population therefore stops declining and grows without bound.

The yellow perch population increases without bound while $w \to 0$

Both species will survive in the long-term if an equilibrium point is reached such that $w, y > 0$.

(h) By considering $\frac{\mathrm{d}w}{\mathrm{d}t}$ and $\frac{\mathrm{d}y}{\mathrm{d}t}$, show that there is no number of yellow perch that could have been introduced to produce such an outcome. [4]

Both populations survive in the long term if the system settles at a point where both w>0 & y>0, Simultaneously.that is, an equilibrium of the coupled system with both populations positive.
Setting

$\frac{dw}{dt} = 0$, we have

$$\frac{w\,(20 - w)}{40} - 10y = 0$$

$$y = \frac{w\,(20 - w)}{400}$$

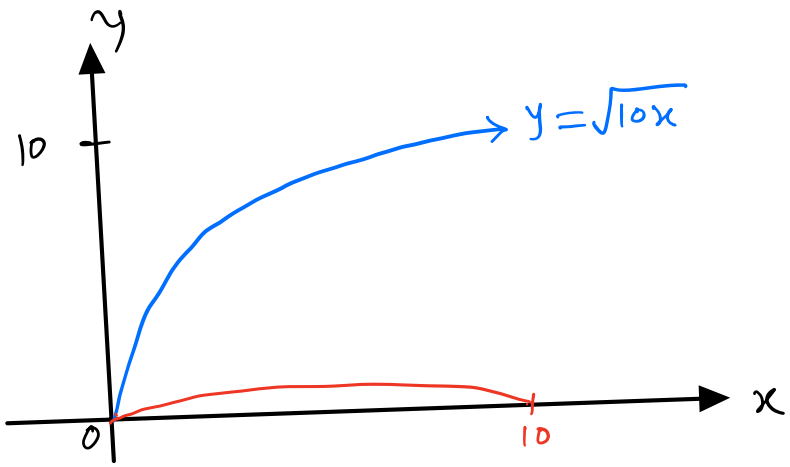
Setting $\frac{dy}{dt} = 0$ and taking positive root, since y represents population, we have

$$\frac{y^2}{10} - w = 0$$

$$y = \sqrt{10w}$$

On GDC, we can sketch the graphs of $y = \frac{x(20-x)}{400}$ and $y = \sqrt{10x}$

Using x in place of w, and any point of intersection other than origin.



The two graphs only meet at the origin. For every w>0, $y = \sqrt{10w}$ lies entirely above $y = \frac{w(20-w)}{400}$

So, there is no equilibrium with both populations positive.

Hence, there is no number of yellow perch that could have been introduced such that a non-zero equilibrium point is reached.

End of Examination

Best of Luck for Exam. If you like my method of teaching then don't forget to LIKE, SHARE & SUBSCRIBE to my YouTube channel.

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